Proofs for "Systematic Unequal-Probability Sampling for Active Statistical Inference and Event Recall under an Uplink Budget on Passive Acoustic Nodes"

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This document contains the assumptions and complete proofs of Lemma 1, Propositions 1 and 2 and Corollary 1 of the paper, of the remarks that the paper makes without proof, and of two supplementary statements: the exact variance difference between the single start and EXCL (Proposition S1) and the sufficiency of the uplink sketch (Proposition S2). Section 7 defines the successive-difference variance estimate used in the supplementary tables. Notation follows the paper. The statements were also checked by exact enumeration and linear programming (directory `enumeration/` of the repository).

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1 Setting and notation

Population and estimator. A node observes units $t = 1, \dots, N$ in time order. Scores s_t (hence bins $j(t)$) and labels $y_t \in \{0, 1\}$ are fixed numbers (design-based view); all randomness comes from the sampling. A rule assigns an upload probability $p_t \in [p_{\min}, 1]$ with $p_{\min} > 0$. The decision is $\xi_t \in \{0, 1\}$, $\mathbb{P}(\xi_t = 1)$ is the inclusion probability, and the target is $\theta = N^{-1} \sum_t y_t$. For a predictor $f_t \in [0, 1]$ put $r_t = y_t - f_t \in [-1, 1]$ and

$$\hat{\theta} = \frac{1}{N} \sum_t \Big[f_t + \xi_t \frac{r_t}{p_t} \Big], \qquad \hat{V} = \frac{1}{N^2} \sum_t \xi_t \frac{r_t^2}{p_t^2}. \tag{1}$$

The interval of the paper is $\hat\theta \pm 1.96\sqrt{\max(\hat V, V_{\text{plug}})}$. Since

$$\hat\theta - \theta = \frac{1}{N}\sum_t r_t\Big(\frac{\xi_t}{p_t} - 1\Big), \tag{2}$$

the error of the estimator is the sum of the centred terms $r_t(\xi_t/p_t - 1)$.

EXCL (blocked restart). Time is cut into blocks $B_1, B_2, \dots$ of consecutive units. Let A_t be the cumulative probability within the block up to and including unit t ($A = 0$ before the first unit of the block). A unit t opens a new block if $A_{t-1} + p_t > 1$, if it starts a recording, or if the onset-alignment condition holds; hence every block has mass $\mu_k = \sum_{t\in B_k} p_t \le 1$. Block k draws one $u_k \sim U(0,1]$ and

$$\xi_t = \mathbf{1}\{A_{t-1} < u_k \le A_t\} \qquad (t \in B_k).$$

The u_k are independent. Let $\mathcal{F}_k = \sigma(u_1, \dots, u_{k-1})$. The number $K \le N$ of blocks may be random; terms with $k > K$ are zero.

Single start (Madow). A *recording* σ is a maximal run of consecutive units without a recording start inside. With $C_t = \sum_{s\le t,\, s\in\sigma} p_s$ (the cumulative probability within the recording, $C = 0$ before its first unit), $I_t = (C_{t-1}, C_t]$ and M_σ the value of C at the last unit, the design draws one $u_\sigma \sim U(0,1]$ per recording, independently across recordings, and

$$\xi_t = 1 \iff (u_\sigma + \mathbb{Z}) \cap I_t \neq \emptyset.$$

Since $|I_t| = p_t \le 1$, at most one point of $u_\sigma + \mathbb{Z}$ lies in I_t. A design is *open loop* if all p_t are functions of the scores alone (not of any draw), and *closed loop* if p_t also depends on earlier decisions.

Events and recall. An event e is a fixed set of units; it is hit if $\sum_{t\in e} \xi_t \ge 1$. Event recall is the mean of $\mathbb{P}(e \text{ hit})$ over events. The mass of e in block B_k is $M_k = \sum_{t\in e\cap B_k} p_t$.

Ordered pairs. Sums over $s \neq t$ run over ordered pairs.

2 Lemma 1: validity of EXCL

Lemma 1 (Validity of EXCL; paper, Lemma 1)**.** *Suppose that the first unit of block k, the extent of B_k and, for $t \in B_k$, p_t and f_t are $\mathcal{F}_k$-measurable (they may use all scores but only draws of earlier blocks). Then*

(i) $\mathbb{P}(\xi_t = 1 \mid \mathcal{F}_k) = p_t$ *for* $t \in B_k$*;*

(ii) $\mathbb{E}[\hat\theta] = \theta$*;*

(iii) $\mathbb{E}[\hat V] - \mathrm{Var}(\hat\theta) = N^{-2}\sum_k \mathbb{E}\big[(\sum_{t\in B_k} r_t)^2\big] \ge 0.$

Proof. (i) Given $\mathcal{F}_k$, the numbers $A_{t-1} < A_t \le 1$ are fixed, with $A_t - A_{t-1} = p_t$ for $t \in B_k$ (the block mass is at most one by the rule that opens blocks, and the extent and p_t are known). The variable $u_k \sim U(0,1]$ is independent of $\mathcal{F}_k$, so $\mathbb{P}(\xi_t = 1 \mid \mathcal{F}_k) = \mathbb{P}(A_{t-1} < u_k \le A_t \mid \mathcal{F}_k) = A_t - A_{t-1} = p_t$.

(ii) Write $R_k = \sum_{t\in B_k} r_t$, $T_k = \sum_{t\in B_k} \xi_t r_t/p_t$ and $Z_k = T_k - R_k = \sum_{t\in B_k} r_t(\xi_t/p_t - 1)$. By (2), $N(\hat\theta - \theta) = \sum_{k\le K} Z_k$. Each block has at most $1/p_{\min}$ units (mass ≤ 1, $p_t \ge p_{\min}$), at most one unit

of a block is uploaded, and $|r_t| \le 1$; hence $|T_k| \le 1/p_{\min}$, $|R_k| \le 1/p_{\min}$ and $|Z_k| \le 2/p_{\min}$. The variable Z_k is $\mathcal{F}_{k+1}$-measurable, and by (i) and the $\mathcal{F}_k$-measurability of r_t and p_t,

$$\mathbb{E}[Z_k \mid \mathcal{F}_k] = \sum_{t \in B_k} r_t \Big(\frac{\mathbb{P}(\xi_t = 1 \mid \mathcal{F}_k)}{p_t} - 1 \Big) = 0.$$

(The event $\{k \le K\}$ is $\mathcal{F}_k$-measurable.) Hence (Z_k) is a bounded martingale-difference sequence and $\mathbb{E}[\hat{\theta}] = \theta$.

(iii) Because the Z_k are bounded martingale differences, $\mathbb{E}[Z_j Z_k] = \mathbb{E}[Z_j \mathbb{E}[Z_k \mid \mathcal{F}_k]] = 0$ for $j < k$, and since $\hat{\theta}$ is unbiased,

$$N^2 \mathrm{Var}(\hat{\theta}) = \sum_k \mathbb{E}[Z_k^2] = \sum_k \mathbb{E}\big[\mathbb{E}[T_k^2 \mid \mathcal{F}_k] - R_k^2\big],$$

where we used $\mathbb{E}[T_k \mid \mathcal{F}_k] = R_k$ (from the display above). The intervals $(A_{t-1}, A_t]$, $t \in B_k$, are disjoint, so at most one ξ_t in a block is non-zero and $T_k^2 = \sum_{t \in B_k} \xi_t r_t^2 / p_t^2$. Therefore $\mathbb{E}[T_k^2 \mid \mathcal{F}_k] = \sum_{t \in B_k} r_t^2 / p_t$ and

$$N^2 \mathbb{E}[\hat{V}] = \mathbb{E} \sum_k \sum_{t \in B_k} \xi_t \frac{r_t^2}{p_t^2} = \mathbb{E} \sum_k \mathbb{E}[T_k^2 \mid \mathcal{F}_k].$$

Subtracting gives $N^2(\mathbb{E}\hat{V} - \mathrm{Var}\hat{\theta}) = \mathbb{E} \sum_k R_k^2 \ge 0$. □

Remark 1 (Conditions used). The proof uses only: the probabilities and the predictor of a block are fixed by the scores and the draws of earlier blocks; the mass of a block is at most one; u_k is uniform and independent of the past. In particular, the price and the token bucket of the closed-loop rule may change only when a block closes, and the periods of the prequential predictor must be aligned to block starts. The estimator $\hat{V}$ in Lemma 1 is the raw one; replacing it by $\max(\hat{V}, V_{\mathrm{plug}})$ can only widen the interval.

Asymptotic conservativeness

The paper states that, with a growing number of blocks, the martingale central limit theorem makes the interval asymptotically conservative. We state the conditions used. Consider a sequence of populations and designs indexed by N for which Lemma 1 applies, with $p_{\min} > 0$ not depending on N, and write $s_N^2 = N^2 \mathrm{Var}(\hat{\theta}) = \sum_k \mathbb{E} Z_k^2$ and $\mathcal{V}_N^2 = \sum_k \mathbb{E}[Z_k^2 \mid \mathcal{F}_k]$.

(A1) (non-degeneracy) $\liminf_N s_N^2 / N > 0$;

(A2) (stability) $\mathcal{V}_N^2 / s_N^2 \to 1$ in probability.

Proposition A 1 (Asymptotic conservativeness)**.** *Under (A1)–(A2),* $(\hat{\theta} - \theta)/\sqrt{\mathrm{Var}\hat{\theta}} \Rightarrow \mathcal{N}(0,1)$ *and* $\liminf_N \mathbb{P}\big(|\hat{\theta} - \theta| \le z\sqrt{\hat{V}}\big) \ge \mathbb{P}(|\mathcal{N}(0,1)| \le z)$ *for every* $z > 0$*; the same holds for the interval with* $\max(\hat{V}, V_{\mathrm{plug}})$.

Proof. Put $X_{N,k} = Z_k / s_N$. These are martingale differences with $|X_{N,k}| \le 2/(p_{\min} s_N) \to 0$ (by (A1), $s_N^2 \to \infty$), so the Lindeberg condition holds; by (A2) the conditional variances sum to $\mathcal{V}_N^2 / s_N^2 \to 1$. The martingale central limit theorem (e.g. [1, Cor. 3.1]) gives the first claim.

For $\hat{V}$: $N^2 \hat{V} = \sum_k T_k^2 = \sum_k \mathbb{E}[T_k^2 \mid \mathcal{F}_k] + D_N$ with $D_N = \sum_k (T_k^2 - \mathbb{E}[T_k^2 \mid \mathcal{F}_k])$. The summands of D_N are martingale differences bounded by $p_{\min}^{-2}$, and there are at most N of them, so $\mathbb{E} D_N^2 \le N p_{\min}^{-4}$ and $D_N / s_N^2 \to 0$ in probability by (A1) and Chebyshev's inequality. Moreover $\mathbb{E}[T_k^2 \mid \mathcal{F}_k] = \mathbb{E}[Z_k^2 \mid \mathcal{F}_k] + R_k^2 \ge \mathbb{E}[Z_k^2 \mid \mathcal{F}_k]$, whence $N^2 \hat{V} \ge \mathcal{V}_N^2 + D_N$ and, by (A2), $N^2 \hat{V} / s_N^2 \ge 1 - o_p(1)$. Fix $\epsilon > 0$; with probability tending to one $\hat{V} \ge (1 - \epsilon)\mathrm{Var}(\hat{\theta})$, so $\liminf \mathbb{P}(|\hat{\theta} - \theta| \le z\sqrt{\hat{V}}) \ge \mathbb{P}(|\mathcal{N}(0,1)| \le z\sqrt{1-\epsilon})$ and $\epsilon \downarrow 0$ gives the claim. Enlarging $\hat{V}$ to $\max(\hat{V}, V_{\mathrm{plug}})$ only enlarges the interval. □

Remark 2. (A1) and (A2) are conditions on the sequence of populations; they are not implied by Lemma 1. The statement concerns the design-based asymptotics of a sequence of populations; the coverage in the tables of the paper is an empirical Monte-Carlo quantity.

3 Proposition 1: the restart cost is the price of conditional marginals

The class $\mathcal{C}$. Units are decided in time order and every decision is final when it is made (*immediate decisions*). Let $\mathcal{F}_s = \sigma(\xi_1, \dots, \xi_s)$ (together with any randomisation used up to s) and let $B_1, B_2, \dots$ be consecutive blocks with first units τ_k. Put $\mathcal{H}_k = \mathcal{F}_{\tau_k - 1}$ (the stopped σ-algebra: all decisions before the block) and, for a unit t, $\mathcal{H}(t) = \mathcal{H}_k$ on $\{t \in B_k\}$. A design is in $\mathcal{C}$ if

(C1) decisions are immediate, so $\mathcal{H}_k$ contains every decision made before B_k;

(C2) the extent of B_k and p_t, $t \in B_k$, are $\mathcal{H}_k$-measurable (functions of the fixed scores and of earlier decisions), and $p_t > 0$; the design does not use the labels y;

(C3) (validity) $\mathbb{E}[\hat{\theta}_f] = \theta$ for every $y \in \{0,1\}^N$ and every predictor f with values in $[0,1]$ whose f_t is $\mathcal{H}(t)$-measurable.

The predictors fitted on uploads of earlier blocks, such as the prequential predictor of the paper, are members of this class. EXCL belongs to $\mathcal{C}$ by Lemma 1(i)–(ii). For an event e let $N_k = \sum_{t \in e \cap B_k} \xi_t$, $M_k = \sum_{t \in e \cap B_k} p_t$ and $D_k = \mathbf{1}\{\xi_s = 0 \ \forall s \in e,\ s < \tau_k\}$, which is $\mathcal{H}_k$-measurable by (C1)–(C2); M_k is $\mathcal{H}_k$-measurable by (C2).

Proposition 1 (paper, Proposition 1)**.** *Let a design satisfy (C1)–(C2).*

(a) (C3) holds if and only if $\mathbb{P}(\xi_t = 1 \mid \mathcal{H}(t)) = p_t$ for all t.

(b) For every design in $\mathcal{C}$ and every event e,

$$\mathbb{E}\sum_k M_k - \mathbb{P}(e \text{ hit}) = \underbrace{\mathbb{E}\sum_k (1 - D_k) M_k}_{\text{restart cost}} + \underbrace{W_e}_{\geq 0}, \qquad W_e = \mathbb{E}\sum_k D_k (N_k - 1)^+,$$

where $1 - D_k = \mathbf{1}\{e \text{ hit before } B_k\}$ and W_e counts repeated hits within blocks.

(c) If the masses $M_k = m_k$ are deterministic (the blocks may otherwise be random, and so may the units of e in them), then every design in $\mathcal{C}$ satisfies $\mathbb{P}(e \text{ hit}) \leq U_e := 1 - \prod_k (1 - \min(1, m_k))$. For EXCL blocks $m_k \leq 1$, so $U_e = 1 - \prod_k (1 - m_k)$.

(d) EXCL has $W_e = 0$ for every event simultaneously. If it is open loop (blocks and p_t determined by the scores), it attains $U_e = 1 - \prod_k (1 - m_k)$ for every event at once. When all $m_k < 1$, a design in $\mathcal{C}$ attains U_e if and only if $W_e = 0$.

Proof. (a) "⇐": by (2), for $t \in B_k$ the factors r_t and p_t are $\mathcal{H}_k$-measurable, so $\mathbb{E}[r_t \xi_t / p_t] = \mathbb{E}[(r_t/p_t)\mathbb{P}(\xi_t = 1 \mid \mathcal{H}(t))] = \mathbb{E}[r_t]$, and (2) gives $\mathbb{E}\hat{\theta}_f = \theta$ for every admissible f (all variables are bounded because $p_t > 0$ takes finitely many values).

"⇒": fix t and $A \in \mathcal{H}(t)$. Take the population with $y_t = 1$ and y_s arbitrary for $s \neq t$ (say 0), the predictor $f_s = y_s$ for $s \neq t$ (a constant, hence admissible) and $f_t = 1 - \mathbf{1}_A$, which is $\mathcal{H}(t)$-measurable with values in $\{0,1\}$. Then $r_s = 0$ for $s \neq t$, $r_t = \mathbf{1}_A$ and (2) gives $0 = \mathbb{E}[\hat{\theta}_f - \theta] = N^{-1}\mathbb{E}[\mathbf{1}_A(\xi_t/p_t - 1)]$. As $A \in \mathcal{H}(t)$ is arbitrary and p_t is $\mathcal{H}(t)$-measurable and positive, $\mathbb{P}(\xi_t = 1 \mid \mathcal{H}(t)) = p_t$. Only $\{0,1\}$-valued predictors were needed.

(b) The first block in which e is hit is unique, so $\mathbf{1}\{e \text{ hit}\} = \sum_k D_k \mathbf{1}\{N_k \geq 1\}$, and $\mathbf{1}\{N \geq 1\} = N - (N-1)^+$ for integer $N \geq 0$. By (a), $\mathbb{E}[N_k \mid \mathcal{H}_k] = M_k$, and D_k is $\mathcal{H}_k$-measurable, so $\mathbb{E}[D_k N_k] = \mathbb{E}[D_k M_k]$. Hence

$$\mathbb{P}(e \text{ hit}) = \mathbb{E} \sum_k D_k M_k - \mathbb{E} \sum_k D_k (N_k - 1)^+,$$

and $\mathbb{E}\sum_k M_k - \mathbb{P}(e \text{ hit}) = \mathbb{E}\sum_k (1 - D_k) M_k + W_e$. Both terms are expectations of non-negative variables. (All sums are finite since there are at most N blocks.)

(c) Index the blocks with $m_k > 0$ by k (blocks with $m_k = 0$ do not meet e and are skipped). Since $D_{k+1} = D_k \mathbf{1}\{N_k = 0\}$ and D_k is $\mathcal{H}_k$-measurable, Markov's inequality and (a), $\mathbb{E}[N_k \mid \mathcal{H}_k] = m_k$ give $\mathbb{P}(N_k \geq 1 \mid \mathcal{H}_k) \leq \min(1, m_k)$ and

$$\mathbb{P}(D_{k+1} = 1) = \mathbb{E}\big[D_k \mathbb{P}(N_k = 0 \mid \mathcal{H}_k)\big] \geq (1 - \min(1, m_k)) \mathbb{P}(D_k = 1).$$

By induction $\mathbb{P}(e \text{ missed}) \geq \prod_k (1 - \min(1, m_k))$.

(d) The intervals $(A_{t-1}, A_t]$, $t \in B_k$, are disjoint and u_k is a single draw, so $N_k \leq 1$ for every event: $W_e = 0$ simultaneously for all e, and by (b) $\mathbb{P}(e \text{ hit}) = \mathbb{E}\sum_k D_k M_k$ (closed loop included). If blocks and p_t are deterministic, $N_k = \mathbf{1}\{u_k \in \bigcup_{t \in e \cap B_k} (A_{t-1}, A_t]\}$ is a Bernoulli(m_k) variable, independent across blocks, so $\mathbb{P}(e \text{ missed}) = \prod_k (1 - m_k)$ and (c) holds with equality.

For the last claim let all $m_k < 1$. If $W_e = 0$ then $N_k \leq 1$ a.s. on $\{D_k = 1\}$, so $\mathbb{P}(N_k \geq 1 \mid \mathcal{H}_k) = \mathbb{E}[N_k \mid \mathcal{H}_k] = m_k$ there and every step of (c) is an equality. Conversely, if U_e is attained, then since $\mathbb{P}(D_{k+1} = 1) \geq (1 - m_k)\mathbb{P}(D_k = 1)$ for every k and the product of the factors is attained by $\mathbb{P}(\text{missed})$, each of these inequalities is an equality; hence $\mathbb{E}[D_k(\mathbb{P}(N_k = 0 \mid \mathcal{H}_k) - (1 - m_k))] = 0$ with a non-negative integrand, i.e. $\mathbb{P}(N_k \geq 1 \mid \mathcal{H}_k) = \mathbb{E}[N_k \mid \mathcal{H}_k]$ a.s. on $\{D_k = 1\}$, which means $\mathbb{E}[(N_k - 1)^+ \mid \mathcal{H}_k] = 0$ there and $W_e = 0$. □

Remark 3 (Corrections to a stronger formulation). (1) If some $m_k = 1$, "attains $U_e \Rightarrow W_e = 0$" is false: take $B_1 = \{a, b\}$ with $p_a = p_b = 1/4$ and independent decisions, $B_2 = \{c\}$ with $p_c = 1$, $e = \{a, b, c\}$; then $m = (1/2, 1)$, $U_e = 1 = \mathbb{P}(e \text{ hit})$ but $W_e = \mathbb{P}(\xi_a = \xi_b = 1) = 1/16$. (2) The bound (c) needs only that the *masses* are deterministic, not the blocks.

The union bound; single start and Deville's design

For any design with unconditional marginals $\mathbb{P}(\xi_t = 1) = p_t$ and any event, Boole's inequality gives $\mathbb{P}(e \text{ hit}) \leq \min(1, \sum_{t \in e} p_t)$ (the *union bound*). For e split over two blocks with $m_1 + m_2 \leq 1$,

$$\min(1, m_1 + m_2) - U_e = m_1 + m_2 - (m_1 + m_2 - m_1 m_2) = m_1 m_2,$$

the loss that every design in $\mathcal{C}$ pays. The following statements delimit the bound.

(a) *Single start.* Take p_t deterministic and one start $u \sim U(0, 1]$ over the cumulative probability axis. Then $\mathbb{P}(\xi_t = 1) = p_t$, and a contiguous event is an interval of length $m_e = \sum_k m_k$ on the circle $\mathbb{R}/\mathbb{Z}$; it is hit iff $u \bmod 1$ lies in this arc, with probability $\min(1, m_e)$, which equals the union bound and exceeds U_e by $m_1 m_2 > 0$ for an event split over two blocks. The design satisfies (C1)–(C2) but violates (C3): given the decisions of earlier blocks it can be deterministic. Example: blocks $\{a\}, \{c\}$ with $p_a = p_c = 1/2$ have $\mathbb{P}(\xi_c = 1 \mid \xi_a = 1) = 0 \neq 1/2$.

(b) *Deville's design.* The systematic (ordered pivotal) design of Deville admits an implementation with immediate decisions [4], so it satisfies (C1)–(C2). By Proposition 1(c), *any* design with (C1)–(C2) and deterministic masses whose hit probability exceeds U_e must violate (C3), i.e.

$\mathbb{P}(\xi_t = 1 \mid \mathcal{H}(t)) \neq p_t$ for some t. In this sense a single start and Deville's design exceed the bound only by giving up the conditional marginals. (The statement does not claim that Deville's design exceeds the bound for every event, only that it can do so only in this way.)

(c) *Delayed decisions.* If (C1) is weakened (decisions of a block are revealed only after the next block has started), the bound need not hold: in the extreme case that nothing is revealed before the end, $\mathcal{H}$ is trivial, admissible predictors are deterministic, and the single start above is valid and beats U_e.

Remark 4 (Closed loop: no uniform optimality). If the masses on e in later blocks depend on earlier outcomes, EXCL is not uniformly optimal. Let $B_1 = \{a, b\}$ with $a \in e$, $b \notin e$, $p_a + p_b \leq 1$, and $B_2 = \{c\}$ with $c \in e$, $p_c = g(\xi_a + \xi_b)$, $g(0) > g(1)$. EXCL gives $\mathbb{P}(\xi_a = 1) = p_a$, $\mathbb{P}(\xi_b = 1) = p_b$ with at most one upload, so $\mathbb{P}(e \text{ hit}) = p_a + p_b g(1) + (1 - p_a - p_b)g(0)$. The design that draws (ξ_a, ξ_b) with $\mathbb{P}(1,1) = x \in (0, \min(p_a, p_b)]$, $\mathbb{P}(1,0) = p_a - x$, $\mathbb{P}(0,1) = p_b - x$ (so that the marginals remain p_a, p_b, and (C3) holds since $\mathcal{H}_1$ is trivial) and $\xi_c \mid \text{past} \sim \text{Bernoulli}(g(\xi_a + \xi_b))$ has $\mathbb{P}(e \text{ hit})$ larger by $x(g(0) - g(1))$, while its expected mass on e differs by $x(g(0) - 2g(1) + g(2))$, which can be negative (for example $p_a = 0.3$, $p_b = 0.4$, $g = (0.9, 0.5, 0)$, $x = 0.25$: hit probability $+0.100$, expected mass -0.025). Hence recall comparisons "at the same p_t" are uniform only in open loop.

4 Corollary 1: the plug-in

Corollary 1 (paper, Corollary 1)**.** *Let p_t be open loop. Then EXCL draws at these probabilities keep p_t, the expected budget $\mathbb{E}\sum_t \xi_t = \sum_t p_t$ and Lemma 1; they hit every event with probability $1 - \prod_k (1 - m_k) \geq 1 - \prod_{t \in e}(1 - p_t)$ (the value of independent draws); and for a fixed predictor they change $N^2 \mathrm{Var}(\hat{\theta})$ by $-\sum_k \sum_{s \neq t \in B_k} r_s r_t$.*

Proof. Since p_t is a function of the scores, the blocks are deterministic, the u_k are independent and Lemma 1 applies. Marginals and budget: Lemma 1(i). Hit probability: Proposition 1(d) gives $1 - \prod_k (1 - m_k)$, with $m_k = \sum_{t \in e \cap B_k} p_t$. For independent draws $\mathbb{P}(e \text{ missed}) = \prod_k a_k$ with $a_k = \prod_{t \in e \cap B_k}(1 - p_t)$, and the Weierstrass product inequality $\prod(1 - p_t) \geq 1 - \sum p_t$ gives $a_k \geq 1 - m_k \geq 0$; therefore $\prod_k a_k \geq \prod_k (1 - m_k)$.

Variance for a fixed predictor: the blocks are independent, $\mathbb{E}T_k = R_k$ and $\mathbb{E}T_k^2 = \sum_{t \in B_k} r_t^2 / p_t$, so $N^2 \mathrm{Var}_{\mathrm{EXCL}}(\hat{\theta}) = \sum_k \mathrm{Var}(T_k) = \sum_t r_t^2 / p_t - \sum_k R_k^2$. For independent draws $N^2 \mathrm{Var}_{\mathrm{ind}}(\hat{\theta}) = \sum_t r_t^2 (1 - p_t)/p_t = \sum_t r_t^2 / p_t - \sum_t r_t^2$. The difference is

$$N^2(\mathrm{Var}_{\mathrm{EXCL}} - \mathrm{Var}_{\mathrm{ind}}) = \sum_t r_t^2 - \sum_k R_k^2 = -\sum_k \sum_{s \neq t \in B_k} r_s r_t.$$

□

Remark 5 (Further facts from the same computation). (1) The variance of the budget is $\mathrm{Var}_{\mathrm{EXCL}}(\sum_t \xi_t) = \sum_k \mu_k(1 - \mu_k)$, against $\mathrm{Var}_{\mathrm{ind}}(\sum_t \xi_t) = \sum_t p_t(1 - p_t)$; the latter is larger by $\sum_k \sum_{s \neq t \in B_k} p_s p_t$ (block counts are Bernoulli(μ_k)). (2) The expected raw variance estimate is never smaller under EXCL: $N^2(\mathbb{E}_{\mathrm{EXCL}} \hat{V} - \mathbb{E}_{\mathrm{ind}} \hat{V}_{\mathrm{ind}}) = \sum_t r_t^2 \geq 0$ for a fixed predictor, where $\hat{V}_{\mathrm{ind}} = N^{-2} \sum_t \xi_t (1 - p_t) r_t^2 / p_t^2$. (3) The change in the variance cannot be estimated without bias from the sample, since pairs in the same block are never uploaded together ($\pi_{st} = 0$). (4) If $\sum_t r_t^2 (1 - p_t)/p_t > 0$, the ratio of standard deviations is $\sqrt{1 - C/S}$ with $C = \sum_k \sum_{s \neq t \in B_k} r_s r_t$ and $S = \sum_t r_t^2 (1 - p_t)/p_t$; whether $C \geq 0$ on the target data is an assumption when only development data are labelled.

5 Proposition 2: open-loop single start

Let p_t be open loop and let the design be the single start of Section 1. On the cumulative-probability axis of a recording σ put

$$\tilde g_\sigma(x) = \frac{r_t}{p_t} \quad (x \in I_t), \qquad \tilde g_\sigma(x) = 0 \quad (x \notin (0, M_\sigma]),$$

and let $\gamma_\sigma(h) = \int_{\mathbb{R}} \tilde g_\sigma(x)\tilde g_\sigma(x+h)\,dx$ be its covariogram (γ_σ is even, continuous, piecewise linear, $\gamma_\sigma(h) = 0$ for $|h| \ge M_\sigma$ and $\gamma_\sigma(0) = \sum_{t\in\sigma} r_t^2/p_t$).

Proposition 2 (paper, Proposition 2)**.** *(i)* $\mathbb{P}(\xi_t = 1) = p_t$; $\mathbb{P}(\xi_s = \xi_t = 1) = \lambda\big((I_s \bmod 1) \cap (I_t \bmod 1)\big)$ *for* s, t *in one recording, and units of different recordings are independent; any set of consecutive units of a recording with mass* m *receives* $\lfloor m\rfloor$ *or* $\lceil m\rceil$ *uploads, in particular a recording receives* $\lfloor M_\sigma\rfloor$ *or* $\lceil M_\sigma\rceil$*; and* $\hat\theta$ *is unbiased for any predictor with* f_t *independent of* $u_{\sigma(t)}$.

(ii) For a fixed predictor,

$$N^2(\mathbb{E}\hat V - \mathrm{Var}\hat\theta) = \sum_\sigma \Delta_\sigma, \qquad \Delta_\sigma = 2\Big[\int_0^\infty \gamma_\sigma(h)\,dh - \sum_{j\ge1}\gamma_\sigma(j)\Big].$$

Thus $\mathbb{E}\hat V \ge \mathrm{Var}\hat\theta$ *iff* $\sum_\sigma \Delta_\sigma \ge 0$*. A recording with* $M_\sigma \le 1$ *has* $\Delta_\sigma = (\sum_{t\in\sigma} r_t)^2 \ge 0$*. If* γ_σ *is nonincreasing and convex on* $[1/2, M_\sigma]$ *and* $\int_0^{1/2}\gamma_\sigma \ge 0$*, then* $\Delta_\sigma \ge 0$*. If* r *is periodic with the sampling interval,* Δ_σ *can be negative without bound relative to* $\mathbb{E}\hat V$.

(iii) With a closed-loop price the marginals are lost: if $p_1 = 0.5$ *and* $p_2 = 0.2$ *when unit 1 was uploaded,* 0.6 *otherwise, then* $\mathbb{P}(\xi_2 = 1) = 1/2$ *and* $\mathbb{E}[\xi_2/p_2] = 0.5/0.6 \approx 0.83$.

Proof. Fix a recording and drop σ.

(i) The variable $u \bmod 1$ is uniform on $\mathbb{R}/\mathbb{Z}$. Because $|I_t| = p_t \le 1$, I_t maps injectively (up to an endpoint) onto an arc $I_t \bmod 1$ of length p_t, and $(u+\mathbb{Z}) \cap I_t \ne \emptyset$ iff $u \bmod 1 \in I_t \bmod 1$. Hence $\mathbb{P}(\xi_t = 1) = p_t$ and $\mathbb{P}(\xi_s = \xi_t = 1) = \lambda((I_s \bmod 1) \cap (I_t \bmod 1))$; different recordings use independent starts.

For consecutive units W of the recording, $\bigcup_{t\in W} I_t = (a, a + m_W]$ for some a (the arcs tile the axis), and each upload in W corresponds to exactly one point of $u+\mathbb{Z}$ in this interval, because at most one point lies in each I_t and the I_t are disjoint. A half-open interval of length m contains $\lfloor m\rfloor$ or $\lceil m\rceil$ points of any translate of $\mathbb{Z}$. If f_t is independent of u, then $\mathbb{E}[\xi_t r_t/p_t] = \mathbb{E}[r_t]$, and (2) gives $\mathbb{E}\hat\theta = \theta$.

(ii) We have $N(\hat\theta - \theta) = \sum_\sigma (T_\sigma - \sum_{t\in\sigma} r_t)$ with $T_\sigma = \sum_{t\in\sigma}\xi_t r_t/p_t$, the recordings are independent, and $N^2\mathbb{E}\hat V = \sum_t r_t^2/p_t = \sum_\sigma \gamma_\sigma(0)$ under every design with marginals p_t. So it suffices to compute $\mathrm{Var}T_\sigma$. Each selected unit receives exactly one point of $u+\mathbb{Z}$, hence $T_\sigma(u) = \sum_{m\in\mathbb{Z}} \tilde g(u+m)$ (points outside $(0, M]$ contribute zero). Because $\tilde g$ has compact support the sums are finite and

$$\mathbb{E}T^2 = \int_0^1 \sum_{m,m'} \tilde g(u+m)\tilde g(u+m')\,du = \sum_{j\in\mathbb{Z}} \int_{\mathbb{R}} \tilde g(x)\tilde g(x+j)\,dx = \sum_{j\in\mathbb{Z}} \gamma(j),$$

$$(\mathbb{E}T)^2 = \Big(\int_{\mathbb{R}} \tilde g\Big)^2 = \iint \tilde g(x)\tilde g(y)\,dx\,dy = \int_{\mathbb{R}} \gamma(h)\,dh.$$

Hence $\mathrm{Var}T = \sum_{j\in\mathbb{Z}}\gamma(j) - \int_{\mathbb{R}}\gamma$ and, using evenness,

$$\gamma(0) - \mathrm{Var}T = \int_{\mathbb{R}}\gamma - 2\sum_{j\ge1}\gamma(j) = 2\Big[\int_0^\infty \gamma - \sum_{j\ge1}\gamma(j)\Big] =: \Delta.$$

If $M \le 1$ then $\gamma(j) = 0$ for $j \ge 1$ and $\Delta = \int_{\mathbb{R}} \gamma = (\int \tilde{g})^2 = (\sum_t r_t)^2$.

Sufficient condition. Suppose γ is nonincreasing and convex on $[1/2, M]$ (so $M > 1/2$) and $\int_0^{1/2} \gamma \ge 0$. Since γ is continuous with $\gamma(M) = 0$, the monotonicity gives $\gamma \ge 0$ on $[1/2, M]$ and a left derivative at M which is ≤ 0, the right derivative being 0; hence γ (extended by 0) is convex on $[1/2, \infty)$. For $j \ge 1$ the interval $[j - \frac{1}{2}, j + \frac{1}{2}] \subset [1/2, \infty)$ and the Hermite–Hadamard inequality for convex functions gives $\gamma(j) \le \int_{j-1/2}^{j+1/2} \gamma$. Summing over $j \ge 1$,

$$\sum_{j \ge 1} \gamma(j) \le \int_{1/2}^{\infty} \gamma = \int_0^{\infty} \gamma - \int_0^{1/2} \gamma \le \int_0^{\infty} \gamma,$$

so $\Delta \ge 0$.

Equal probabilities and the periodic counterexample. Let $p_t \equiv 1/k$ ($k \ge 2$ an integer) on a recording of $n = Lk$ units, so $I_t = ((t-1)/k, t/k]$ and $\tilde{g} = kr_t$ on I_t. A direct computation of the overlaps gives, for $h = i/k + s$ with $0 \le s \le 1/k$, $\gamma(h) = k^2[(1/k - s)c_i + sc_{i+1}]$ with $c_i = \sum_t r_t r_{t+i}$: γ is the piecewise-linear interpolation of the values kc_i at the lags i/k. In this case the condition above becomes a condition on the sequence of (expected) autocovariances (Cochran's condition for systematic sampling [2]), when r is modelled as random and independent of the design. Now let $r_t = a_{t \bmod k}$ with $\sum_\phi a_\phi = 0$ and $a \ne 0$ (periodic with the sampling interval). The phase $\phi = \lceil ku \rceil$ is uniform on $\{1, \dots, k\}$ and the selected units are those with $t \equiv \phi \pmod k$, so $T = k \sum_{j=0}^{L-1} r_{\phi + jk} = kLa_\phi$ and $\sum_t r_t = L \sum_\phi a_\phi = 0$. Therefore

$$N^2 \mathrm{Var}\hat{\theta} = \mathrm{Var}T = \frac{1}{k} \sum_\phi k^2 L^2 a_\phi^2 = kL^2 \sum_\phi a_\phi^2, \qquad N^2 \mathbb{E}\hat{V} = \sum_t \frac{r_t^2}{p_t} = kL \sum_\phi a_\phi^2,$$

so $\mathrm{Var}\hat{\theta}/\mathbb{E}\hat{V} = L \to \infty$ and $\Delta = kL(1-L)\sum_\phi a_\phi^2 < 0$ for $L \ge 2$. EXCL with blocks equal to the consecutive k-tuples has $R_k = \sum_\phi a_\phi = 0$ for every block, hence $\mathbb{E}\hat{V} = \mathrm{Var}\hat{\theta}$ exactly by Lemma 1(iii).

(iii) Unit 1: $I_1 = (0, 1/2]$, so $\xi_1 = \mathbf{1}\{u \le 1/2\}$. If $u \le 1/2$ then $\xi_1 = 1$, $p_2 = 0.2$, $I_2 = (1/2, 0.7]$ and neither u nor $u + 1 \in (1, 1.5]$ lies in I_2, so $\xi_2 = 0$. If $u > 1/2$ then $\xi_1 = 0$, $p_2 = 0.6$, $I_2 = (1/2, 1.1] \ni u$ and $\xi_2 = 1$. Hence $\mathbb{P}(\xi_2 = 1) = 1/2$ and $\mathbb{E}[\xi_2/p_2] = \frac{1}{2} \cdot \frac{1}{0.6} = 5/6$, whereas unbiasedness would require 1. Under EXCL the price is frozen within a block and changes only at block starts, $\mathbb{P}(\xi_t = 1 \mid \text{draws of earlier blocks}) = p_t$ by Lemma 1. □

Remark 6 (Predictors fitted within a recording). Even in the open-loop case the single start does not have exact conditional marginals: $\mathbb{P}(\xi_t = 1 \mid \xi_s,\ s < t) \ne p_t$ in general. Hence unbiasedness can fail for predictors that use earlier uploads of the same recording (such as the prequential predictor within a recording): for one recording with $p = (1/2, 1/2)$, $y = (1, 1)$, $f_1 = 0$, $f_2 = \xi_1 y_1$, one finds $2\hat{\theta} = 3\xi_1 + 2(1 - \xi_1)$ and $\mathbb{E}\hat{\theta} = 5/4 \ne \theta = 1$, whereas the independent draw gives $\mathbb{E}\hat{\theta} = 1$. Part (i) therefore asks that f_t be independent of the start of the recording.

6 Supplementary statements

6.1 Variance comparison between the single start and EXCL

Proposition S 1 (exact variance difference)**.** *Let p_t be open loop, the predictor fixed, and let EXCL use blocks within the recordings with the same p_t (every recording start is a block start). Then within every block the joint law of $(\xi_t)_{t \in B_k}$ is the same under the single start and under EXCL, and*

$$N^2 \big[\mathrm{Var}_{\mathrm{Madow}}(\hat{\theta}) - \mathrm{Var}_{\mathrm{EXCL}}(\hat{\theta})\big] = \Gamma := \sum_{\substack{s \ne t \text{ in one recording} \\ \text{in different blocks}}} \frac{r_s r_t}{p_s p_t} \big(\pi_{st}^{\mathrm{M}} - p_s p_t\big),$$

where $\pi^{\mathrm{M}}_{st} = \lambda((I_s \bmod 1) \cap (I_t \bmod 1))$ *is the joint inclusion probability of the single start. Moreover* $N^2(\mathbb{E}\hat{V} - \mathrm{Var}_{\mathrm{Madow}}\hat{\theta}) = \sum_k R_k^2 - \Gamma$ *with* $R_k = \sum_{t \in B_k} r_t$.

Proof. A block B_k is a set of consecutive units of one recording with $\bigcup_{t \in B_k} I_t = (a_k, a_k + \mu_k]$, $\mu_k \leq 1$. Under the single start, $u + \mathbb{Z}$ has at most one point in this interval, and the position $a_k + v$ of that point, $v = (u - a_k) \bmod 1$, is uniform on $(0, 1]$, the point falling inside iff $v \leq \mu_k$. Under EXCL the point $a_k + u_k$ lies in the block iff $u_k \leq \mu_k$, and u_k is uniform. In both designs there is at most one upload, and unit t is uploaded iff the uniform variable falls in the sub-interval of length p_t belonging to t: the joint law of $(\xi_t)_{t \in B_k}$ is the same, and $\mathrm{Var}_{\mathrm{Madow}}(T_k) = \mathrm{Var}_{\mathrm{EXCL}}(T_k)$ with $T_k = \sum_{t \in B_k} \xi_t r_t / p_t$.

Now $N(\hat{\theta} - \theta) = \sum_k (T_k - R_k)$. Under EXCL the blocks are independent, so $N^2 \mathrm{Var}_{\mathrm{EXCL}} = \sum_k \mathrm{Var}(T_k)$. Under the single start, recordings are independent and

$$N^2 \mathrm{Var}_{\mathrm{Madow}} = \sum_k \mathrm{Var}(T_k) + \sum_{k \neq l \text{ in one recording}} \mathrm{Cov}_{\mathrm{M}}(T_k, T_l),$$

with $\mathrm{Cov}_{\mathrm{M}}(T_k, T_l) = \sum_{s \in B_k, t \in B_l} \frac{r_s r_t}{p_s p_t}(\pi^{\mathrm{M}}_{st} - p_s p_t)$ (the sum over ordered pairs $k \neq l$ counts each pair of blocks twice, matching the ordered pairs $s \neq t$ in Γ). Subtracting gives the first identity. For the second, $N^2 \mathbb{E}\hat{V} = \sum_t r_t^2 / p_t$ (any design with marginals p_t), $N^2 \mathrm{Var}_{\mathrm{EXCL}} = \sum_t r_t^2 / p_t - \sum_k R_k^2$ (proof of Corollary 1), so $N^2(\mathbb{E}\hat{V} - \mathrm{Var}_{\mathrm{Madow}}) = \sum_k R_k^2 - \Gamma$. □

Remark 7. Consequently $\Gamma \leq 0$ implies $\mathrm{Var}_{\mathrm{Madow}} \leq \mathrm{Var}_{\mathrm{EXCL}} \leq \mathbb{E}\hat{V}$; $0 < \Gamma \leq \sum_k R_k^2$ means that the single start is less precise than EXCL but its raw variance estimate is still conservative; and $\Gamma > \sum_k R_k^2$ means that it is anti-conservative. Combining with Proposition 2(ii), $\Gamma = \sum_\sigma \left[2 \sum_{j \geq 1} \gamma_\sigma(j) - 2 \int_0^\infty \gamma_\sigma\right] + \sum_k R_k^2$. The check quantity used in the supplementary tables is $\kappa = N^2 \mathrm{Var}_{\mathrm{Madow}}(\hat{\theta}) / \sum_t r_t^2 / p_t$: by Proposition 2(ii), $N^2(\mathbb{E}\hat{V} - \mathrm{Var}_{\mathrm{Madow}}\hat{\theta}) = (1 - \kappa) \sum_t r_t^2 / p_t$, so $\kappa \leq 1$ iff the raw variance estimate is conservative in expectation. It is computed exactly from the scores, the probabilities and the predictor (script `enumeration/madow_exact.py`); on development data it is a certificate only for the development data, and its transfer to test data is an assumption that the tables check empirically.

6.2 Sufficiency of the uplink sketch

Proposition S 2 (sketch sufficiency)**.** *Let time be cut into periods* $k = 1, 2, \ldots$ *and let each unit* t *have a bin* $j(t)$ *and a probability level* $\ell(t)$*, meaning that its upload probability is* $p_t = p_{\ell(t)}$ *(the quantised value carried by the node). The node sends (a) the counts* $C[k, j, \ell]$ *of all units in period* k *with bin* j *and level* ℓ*, and (b) for every uploaded unit its time stamp (hence period* k_t*), bin, level and label* y_t*. Let the shore predictor depend on a unit only through its (bin, period),* $f_t = g_{k(t)}(j(t))$*, with* g_k *a function of the uploads and counts of periods* $< k$*, and let* $V_{\mathrm{plug}} = N^{-2} \sum_t \tilde{v}_t / p_t$ *with* $\tilde{v}_t = \tilde{v}_{k(t), j(t)}$ *a function of* $g_{k(t)}(j(t))$ *and development statistics. Then* $\hat{\theta}$*,* $\hat{V}$ *and* V_{plug} *are determined exactly by (a) and (b), for every such predictor.*

Proof. The functions g_k are computed from the data of periods $< k$, which are contained in (a) and (b); thus all $g_k(j)$ are determined, including for the periods in which units were not uploaded. Then

$$\sum_t f_t = \sum_{k,j} g_k(j) \sum_\ell C[k, j, \ell], \qquad N = \sum_{k,j,\ell} C[k, j, \ell],$$

$$\sum_t \xi_t \frac{r_t}{p_t} = \sum_{t \text{ uploaded}} \frac{y_t - g_{k_t}(j_t)}{p_{\ell_t}}, \qquad N^2 \hat{V} = \sum_{t \text{ uploaded}} \frac{(y_t - g_{k_t}(j_t))^2}{p_{\ell_t}^2},$$

and $N^2 V_{\mathrm{plug}} = \sum_{k,j,\ell} C[k, j, \ell]\, \tilde{v}_{k,j} / p_\ell$. All right-hand sides use only (a) and (b). Hence the shore can replace the predictor without any downlink. □

Remark 8. The statement is exact arithmetic. It does not need the draws u. The prequential predictor of the paper (intercept a_k of period k fitted to the weighted uploads $1/p$ of earlier periods) is a member of the class. The statement requires that the upload records carry the period (time stamp) and the level, and that g_k does not use scores or labels of non-uploaded units of earlier periods.

7 Successive-difference variance estimate

Let $T_k = \sum_{t \in B_k} \xi_t r_t / p_t$ be the block total. The estimate of the paper is $V_1 = \hat{V} = N^{-2} \sum_k T_k^2$ (because at most one unit is uploaded per block, $\sum_k T_k^2 = \sum_t \xi_t r_t^2 / p_t^2$). The successive-difference estimate uses consecutive blocks *within a recording*. For a recording with blocks $k_1 < \cdots < k_m$ define

$$w_{k_i} = \begin{cases} (T_{k_i} - T_{k_{i+1}})^2, & i < m \quad (m \geq 2), \\ (T_{k_{m-1}} - T_{k_m})^2, & i = m \quad (m \geq 2;\ \text{the last difference is repeated}), \\ 2T_{k_1}^2, & m = 1, \end{cases} \qquad V_2 = \frac{1}{N^2} \cdot \frac{1}{2} \sum_k w_k.$$

Thus a recording with $m \geq 2$ blocks contributes $\frac{1}{2}\big[\sum_{i<m}(T_{k_i} - T_{k_{i+1}})^2 + (T_{k_{m-1}} - T_{k_m})^2\big]$ to $N^2 V_2$ (m squared differences, the last one counted twice), and a recording with a single block contributes $T_{k_1}^2$, its contribution to $N^2 V_1$. The intervals use $\max(V_2, V_{\text{plug}})$ in place of $\max(V_1, V_{\text{plug}})$. This is the successive-difference estimator of systematic and one-per-stratum sampling [3], applied to the block totals.

Remark 9 (Expectation in the open-loop case). Let p_t be open loop and the predictor fixed, so that the blocks are deterministic and the T_k are independent with $\mathbb{E}T_k = R_k$; write $v_k = \mathrm{Var}T_k$ and $\delta_i = R_{k_i} - R_{k_{i+1}}$. For consecutive blocks, $\mathbb{E}(T_{k_i} - T_{k_{i+1}})^2 = v_{k_i} + v_{k_{i+1}} + \delta_i^2$. For a recording with $m \geq 2$ blocks the contribution to $N^2 \mathbb{E}V_2$ is therefore

$$\frac{1}{2}\Big[\sum_{i<m}(v_{k_i} + v_{k_{i+1}} + \delta_i^2) + v_{k_{m-1}} + v_{k_m} + \delta_{m-1}^2\Big] = \sum_{i=1}^{m} v_{k_i} + \frac{1}{2}\big(v_{k_{m-1}} - v_{k_1}\big) + \frac{1}{2}\Big[\sum_{i<m}\delta_i^2 + \delta_{m-1}^2\Big],$$

and for a single-block recording it is $v_{k_1} + R_{k_1}^2$. With $N^2 \mathrm{Var}\hat{\theta} = \sum_k v_k$,

$$N^2(\mathbb{E}V_2 - \mathrm{Var}\hat{\theta}) = \sum_{\text{rec.},\, m \geq 2} \frac{1}{2}\Big[v_{k_{m-1}} - v_{k_1} + \sum_{i<m}\delta_i^2 + \delta_{m-1}^2\Big] + \sum_{\text{rec.},\, m=1} R_{k_1}^2.$$

Compared with $N^2(\mathbb{E}V_1 - \mathrm{Var}\hat{\theta}) = \sum_k R_k^2$, the squared block totals are replaced by squared differences of consecutive block totals, which are small when the residual totals vary slowly. The edge term $\frac{1}{2}(v_{k_{m-1}} - v_{k_1})$ has no fixed sign (it vanishes for $m = 2$), so V_2 is not conservative in expectation for every population; it is one block variance per recording against the sum over all its blocks. In closed loop the blocks are not independent and no identity is claimed. The tables report the empirical ratios $\mathbb{E}V_2 / \mathrm{Var}\hat{\theta}$ and the coverage.

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